Project Progress Report

Classification and Measurement of One-Dimensional Hyperuniform Point Patterns

Michael Fang — Advisor: Prof. Salvatore Torquato, Princeton University

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1 Project Overview

This project provides a systematic numerical characterization of hyperuniform point patterns in one dimension. Two complementary metrics are computed for each pattern:

- **Hyperuniformity exponent** α: characterizes how $S(k) \sim |k|^\alpha$ as $k \to 0$.
- **Surface-area coefficient** $\bar{\Lambda}$: measures residual disorder within the strongest (Class I) patterns; $\bar{\Lambda} = 1/6$ is the global minimum (integer lattice).

Three universality classes arise from the large-R behavior of the number variance $\sigma^2(R)$:

Class	Condition	Variance	$\bar{\Lambda}$
I	$\alpha > 1$	bounded	finite
II	$\alpha = 1$	$\sim C_{II} \ln R$	diverges
III	$0 < \alpha < 1$	$\sim A_{III} R^{1-\alpha}$	diverges

The project is organized by "weeks," each corresponding to a checkpoint meeting with Prof. Torquato (not necessarily one calendar week). Three such meetings have taken place to date, yielding 27 characterized patterns and a Junior Paper (§1–§3 complete, 9 pages; results sections pending).

2 Week 1

2.1 Code Benchmarking and Validation

The core numerical algorithm for computing $\sigma^2(R)$ was validated against the 1D Poisson process ($\sigma^2 = 2\rho R$) to within **1.1% mean relative error** across the full R-range. Output: `fig1_poisson_benchmark.png`.

2.2 Quasicrystal Generation

Two generation methods were implemented:

1. **Substitution method** (`substitution_tilings.py`): iterative rule application up to $N \sim 10^7$ for all five metallic-mean chains.
2. **Cut-and-project method** (`projection_method.py`): Fibonacci chain from $\mathbb{Z}^2$ projection. Agreement with substitution: $< 0.1\%$ on $\bar{\Lambda}$.

Chain	Metallic mean μ_n	$\lvert\det M\rvert$	α (exact)
Fibonacci ($n = 1$)	$(1+\sqrt{5})/2 \approx 1.618$	1	3
Silver ($n = 2$)	$1+\sqrt{2} \approx 2.414$	1	3
Bronze ($n = 3$)	$(3+\sqrt{13})/2 \approx 3.303$	1	3
Copper ($n = 4$)	$2+\sqrt{5} \approx 4.236$	1	3
Nickel ($n = 5$)	$(5+\sqrt{29})/2 \approx 5.193$	1	3

All 2×2 metallic-mean matrices have $|\det M| = 1$, which forces $|\lambda_2| = 1/\lambda_1$ and thus $\alpha = 3$ exactly for all of them.

2.3 Number Variance and $\bar{\Lambda}$

$\sigma^2(R)$ was computed for all five metallic-mean chains at $N \sim 10^6$–10^7. All are Class I (bounded, oscillating variance). The exact formula for the integer lattice is $\sigma^2(R) = \{R\}(1 - 2\{R\})$ where $\{R\}$ denotes the fractional part.

Pattern	$\bar{\Lambda}$	Notes
Integer Lattice	$1/6 = 0.1667$	Literature minimum
Fibonacci	0.2011 ± 0.001	Confirms Zachary & Torquato (2009)
Silver	0.2500	Novel (later confirmed = 1/4 exactly; see §4)
Bronze	0.2821 ± 0.0006	Novel
Copper	0.293	Novel
Nickel	0.310	Novel

2.4 Spreadability

Point patterns are embedded into two-phase media ($\phi_2 = 0.35$). The excess spreadability $E(t) = \mathcal{S}(\infty) - \mathcal{S}(t)$ yields α from the long-time slope $E(t) \sim t^{-(1+\alpha)/2}$.

Pattern	α (measured)	α (expected)	Error
Fibonacci	3.049	3	1.6%
Silver	2.992	3	0.3%
Bronze	2.987	3	0.4%

Integer lattice confirmed $\alpha \to \infty$ (exponential decay). Spreadability was validated as a second, independent measure of α.

3 Week 2

3.1 Uncorrelated Random Lattice and Stable Distributions

The uncorrelated random lattice (URL) displaces each site by $U(-a, a)$. The exact formula $\bar{\Lambda}(a) = (1 + a^2)/6$ was derived and validated to $< 0.3\%$ for all a. URL gives $\alpha = 2$ for all $a > 0$ (Class I). At $a = 1$ (cloaking limit), $\bar{\Lambda} = 1/3$ exactly — this turns out to be the limit of the metallic-mean sequence.

Symmetric α-stable perturbations fill the $1 < \alpha < 2$ space stochastically (Class II/III for stability index $s < 1$).

3.2 Stealthy Hyperuniform Patterns

Stealthy configurations ($S(k) = 0$ for $|k| < K^*$) were analyzed:

- Collaborator-provided ensemble: ~4300 disordered-phase configurations per χ ($N = 2000$, $\rho = 1$, $\chi \in \{0.1, 0.2, 0.3\}$).
- 1D approximation derived: $\bar{\Lambda} \approx 1/(\pi^2\chi)$, validated below.

χ	$\bar{\Lambda}$ (measured)	$\bar{\Lambda} = 1/(\pi^2\chi)$	Excess
0.10	1.021	1.013	+0.8%
0.20	0.526	0.507	+3.7%
0.30	0.357	0.338	+5.6%

Two-phase discovery: Generating stealthy configurations via collective-coordinate optimization from a *perturbed lattice* starting point gives $\bar{\Lambda} \approx 1/6$ — near the lattice minimum — while collaborator data (random starting positions) gives $\bar{\Lambda} \approx 1/(\pi^2\chi) \gg 1/6$. Both phases satisfy $S(k) = 0$ for $|k| < K^*$ but differ drastically in their spatial structure. This shows stealthy hyperuniformity does not uniquely determine a point pattern.

Re-classification: Stealthy patterns are not power-law suppressed; α is not meaningful. The correct descriptor is $K^* = 4\pi\rho\chi$.

3.3 New Substitution Chains

Three new chains were added to `substitution_tilings.py`:

Chain	Rules	Matrix	Class	α
Period-Doubling	$S \to L$, $L \to LSS$	$[[0,2],[1,1]]$	II	1.000
0222 Chain	$S \to LL$, $L \to SSLL$	$[[0,2],[2,2]]$	III	0.639
Bombieri-Taylor	$a \to aac$, $b \to ac$, $c \to b$	3×3	I	1.545

Bombieri-Taylor fills the $1 < \alpha < 2$ gap:

$$M_{BT} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \lambda_1 = 2.247, \quad |\lambda_2| = 0.802, \quad \alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = \mathbf{1.545}, \quad \bar{\Lambda} = 0.377$$

This is the first known Class I quasicrystal with $1 < \alpha < 2$.

A new $\alpha = 2$ chain was also added (first deterministic substitution tiling at this value, previously only the stochastic URL occupied $\alpha = 2$):

$$a \to bc,\ b \to c,\ c \to abc \quad \Rightarrow \quad \lambda_1 \approx 1.839,\ |\lambda_2| \approx 0.737,\ \alpha = 2.000,\ \bar{\Lambda} = 0.275.$$

3.4 Full Catalog Assembly and Junior Paper

All patterns were compiled into `catalog.json` via `research_catalog.py`. Infrastructure: `disordered_patterns.p` `stealthy_analysis.py`, `make_figures.py`.

The Junior Paper (`jp/jp_hyperuniformity.tex`) was drafted: §1 Introduction, §2 Theoretical Background, §3 Methods (5 subsections). 9 pages, biblatex+biber backend. Results sections (§4–§8) exist as a separate draft (`jp/jp_results_draft.tex`) pending organization.

4 Week 3

The following work was completed between the second and third meetings.

4.1 Bombieri-Taylor Deep Dive

Script: `bt_deep_dive.py` **Figure:** `fig_bt_deep.png`

1. BT has **32,517 Bragg peaks** in $k < 50$ vs. 2,637 for Fibonacci (12× denser).
2. **Rod overlap:** minimum inter-point spacing $= 1.000 < 2a = 1.002$; correcting a changes the form factor.
3. **Slow convergence:** $|\lambda_2| = 0.802$ creates log-periodic oscillations with period $2\ln\theta_1 = 1.619$. At $t \in [10^5, 10^9]$: $\hat{\alpha} = 1.67$ (theory: 1.545). Converges extremely slowly from above.

Conclusion: Spreadability is unreliable for BT due to rod overlap and large $|\lambda_2|$. Use variance-based $\bar{\Lambda} = 0.377$ instead.

4.2 Finite-Size Scaling of α

Script: `finite_size_scaling.py` **Figure:** `fig_finite_size_alpha_landscape.png`

Direct $S(k)$ power-law fitting fails for all substitution tilings (structure factor dominated by Bragg peaks). Variance-based metrics are reliable:

Pattern	**Class**	**Metric**	$N = 1000$	$N = 10^6$
Bombieri-Taylor	I	$\bar{\Lambda}(N)$	0.3775	0.3783 (stable from $N = 2000$)
Fibonacci	I	$\bar{\Lambda}(N)$	0.1994	0.1982 (stable from $N = 1600$)
Period-Doubling	II	$C_{II}(N)$	0.1284	0.1233 (needs $N > 10^5$)
0222 Chain	III	$\hat{\alpha}(N)$	1.046	0.776 (slow; theory = 0.639)

The 0222 chain is slow because $|\lambda_2| = 1.236 > 1$ (not Pisot), causing large finite-size corrections even at $N = 1.5 \times 10^6$.

4.3 Systematic Search for New Substitution Chains

Script: `search_cubic_tilings.py` **Figure:** `fig_new_tilings.png`

Enumerated all 39,304 3×3 non-negative integer matrices with row sums ≤ 4. 4,882 pass the Pisot condition; 305 have nominal $\alpha \in (2, 3)$.

Result: All 305 give $\alpha = 2.000$ or $\alpha = 3.000$ exactly. No matrix fills $2 < \alpha < 3$.

Proof: For 3×3 unimodular with complex conjugate pair $\lambda_2 = \bar{\lambda}_3$:

$$|\det M| = 1 \Rightarrow \lambda_1 |\lambda_2|^2 = 1 \Rightarrow |\lambda_2| = \lambda_1^{-1/2} \Rightarrow \alpha = 2 \text{ exactly.}$$

Extended to 4×4: Search of 24M matrices (1.36M Pisot) confirms zero cases with $\alpha \in (2, 3)$ among unimodular matrices. Rank-≥ 4 unimodular matrices with complex pairs give $\alpha < 2$ (additional real Pisot factor λ_4 forces $|\lambda_2|^2 < 1/\lambda_1$), so the conclusion holds for any rank.

4.4 Generalized Scaling Metrics for Classes II and III

Script: `generalized_metrics.py` **Figure:** `fig_generalized_metrics.png`

$$C_{II} = \lim_{R \to \infty} \frac{\sigma^2(R)}{\ln R} \qquad \text{(Class II)}$$

$$A_{III} = \lim_{R \to \infty} \frac{\sigma^2(R)}{R^{1-\alpha}} \qquad \text{(Class III)}$$

Results: Period-Doubling $C_{II} = 0.123 \pm 0.016$; 0222 Chain $A_{III} = 0.151 \pm 0.014$ (using $\alpha = 0.639$). Function `compute_generalized_metric()` added to `quasicrystal_variance.py`.

4.5 $\bar{\Lambda}$ Finite-N Corrections

Script: `finite_size_lambda_bar.py` **Figure:** `fig_finite_size_lambda.png`

Integer lattice: The piecewise-cubic integral evaluates to $\bar{\Lambda} = 1/6$ exactly for all integer $R_{\max}$. No finite-N correction exists; observed deviations ($< 0.3\%$) are Monte Carlo noise.

Stealthy: Ordered-phase stealthy (lattice start) gives $\bar{\Lambda} \approx 1/6$ at all N and χ, not $1/(\pi^2 \chi)$. This confirms the two-phase picture: both phases are stealthy but differ drastically in spatial structure.

4.6 Catalog Cleanup and Density Normalization

Script: `catalog_cleanup.py` **Figure:** `fig_catalog_updated.png`

Stealthy patterns relabeled: α removed, $K^* = 4\pi\rho\chi$ added.

$\bar{\Lambda}$ is invariant under uniform density rescaling in $d = 1$: rescaling $x_i \to x_i/\rho$ leaves $\bar{\Lambda}$ unchanged ($< 0.2\%$ numerical error on Fibonacci). Direct comparison of $\bar{\Lambda}$ across all patterns with different densities is valid.

4.7 The $2 < \alpha < 3$ Gap: Non-Unimodular Extension

Scripts: `search_4x4_tilings.py`, `fig_nonunimodular_chain.py` **Figure:** `fig_nonunimodular.png`

The determinant constraint for all 2×2 Pisot matrices gives a unified formula:

$$|\lambda_2| = \frac{|\det M|}{\lambda_1} \implies \alpha = 3 - \frac{2\ln|\det M|}{\ln \lambda_1}.$$

For $|\det M| = 1$ (unimodular): $\alpha = 3$. For $1 < |\det M| < \sqrt{\lambda_1}$: $\alpha \in (2, 3)$.

A targeted search over 2×2 matrices with row sums ≤ 6 found **28 matrices with** $\alpha \in (2, 3)$, all with $|\det M| = 2$. Minimum required row sum is 6 (one rule must produce at least 6 tiles).

Smallest example: $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ ($\det = -2$), rules $a \to b$, $b \to aabbbb$.

$$\lambda_1 = 2 + \sqrt{6} \approx 4.449, \quad |\lambda_2| = |2 - \sqrt{6}| \approx 0.449, \quad \alpha = 2.071.$$

Simulated at $N_{\text{iter}} = 9$: $N = 202{,}256$ points, $\rho = 0.2959$. $\sigma^2(R)$ shows **large log-periodic oscillations** over the range $[0.001, 0.87]$ with log-period $\ln \lambda_1 \approx 1.493$ in $\ln R$. Running average gives $\bar{\Lambda} = 0.650 \pm 0.015$. This is the **first confirmed 1D substitution tiling with** $\alpha \in (2, 3)$.

4.8 High-Precision Silver $\bar{\Lambda}$ and the Metallic-Mean Limit

Scripts: `silver_precision_analysis.py`, `metallic_mean_convergence.py` **Figure:** `fig_metallic_converge`

Silver: High-precision run at $N = 3.88 \times 10^6$ (3×10^4 windows, 1500 R values) gives $\bar{\Lambda} = 0.2501 \pm 0.0004$, which is 0.3σ from $1/4$. The Silver chain has $\rho = 1/2$ exactly (mean spacing $= 2$). The result $\bar{\Lambda} = 1/4$ appears to be exact (novel; analytical proof not yet known).

Metallic-mean limit: $\bar{\Lambda}(\mu_n)$ was computed for $n = 1 \ldots 20$:

n	1	2	3	4	5	6	7	8	10
$\bar{\Lambda}$	0.2011	0.2500	0.2821	0.293	0.310	0.3177	0.3212	0.3249	0.3265
n	15	20	∞						
$\bar{\Lambda}$	0.3289	0.3310	1/3						

Convergence rate: $1/3 - \bar{\Lambda}(n) \approx 0.22/n^{1.43}$. The URL cloaked lattice ($a = 1$, $\bar{\Lambda} = 1/3$ exactly) is the limit of this family.

5 Complete Pattern Catalog

Patterns sorted by $\bar{\Lambda}$ within Class I.

Pattern	**Class**	$\boldsymbol{\alpha}$	$\bar{\boldsymbol{\Lambda}}$
Integer Lattice	I	∞	$1/6 = 0.167$
URL ($a = 0.1$)	I	2.0	0.168
URL ($a = 0.3$)	I	2.0	0.181
Gaussian $\sigma = 0.1$	I	2.0	0.187
Fibonacci	I	3.0	0.201
URL ($a = 0.5$)	I	2.0	0.208
Silver	I	3.0	$\mathbf{1/4} = 0.250$
Cubic α=2 (new)	I	2.0	0.275
Bronze	I	3.0	0.282
Copper	I	3.0	0.293
Nickel	I	3.0	0.310
URL ($a = 1.0$)	I	2.0	0.333
Stealthy ($\chi = 0.3$)	I	—†	0.358
Bombieri-Taylor	I	1.545	0.377
Stable $s = 1.7$	I	1.7	0.242
Stable $s = 1.5$	I	1.5	0.303
Stable $s = 1.3$	I	1.3	0.417
Stealthy ($\chi = 0.2$)	I	—†	0.526
Stealthy ($\chi = 0.1$)	I	—†	1.021
Period-Doubling	II	1.0	div. ($C_{II} = 0.123$)
0222 Chain	III	0.639	div. ($A_{III} = 0.151$)
Cauchy $\gamma = 0.1$	II	0.95	div.
Stable $s = 0.7$	III	0.7	div.
Stable $s = 0.5$	III	0.5	div.
Stable $s = 0.3$	III	0.3	div.

† Stealthy: $S(k) = 0$ by construction; α not applicable. Green rows: literature-confirmed benchmarks. Yellow: novel additions.

6 Figures and Scripts

Figures

Filename	Content
`fig1_poisson_benchmark.png`	Poisson variance validation
`fig2_bounded_variance.png`	$\sigma^2(R)$ for Fibonacci/Silver/Bronze
`fig3_hyperuniformity_test.png`	Normalized variance across classes
`fig9_stealthy_sk.png`	$S(k)$ for stealthy at $\chi = 0.1, 0.2, 0.3$
`stealthy_Sk_overlay.png`	$S(k)$ overlay of all stealthy configs
`bombieri_taylor_analysis.png`	BT variance, $S(k)$, spreadability
`fig12_ranking.png`	Full $\bar{\Lambda}$ ranking, Class I
`fig_variance_II_III.png`	$\sigma^2(R)$ for Period-Doubling and 0222
`fig_bt_deep.png`	BT oscillations, Bragg peak density, $E(t)$
`fig_finite_size_alpha.png`	$\bar{\Lambda}(N)$, $C_{II}(N)$, $\hat{\alpha}(N)$ (portrait)
`fig_finite_size_alpha_landscape.png`	Same, landscape 1×4 for slides
`fig_new_tilings.png`	Cubic α=2 chain variance and $S(k)$
`fig_generalized_metrics.png`	Normalized variance plateaus (Class II/III)
`fig_finite_size_lambda.png`	Lattice $\bar{\Lambda}$ analytic vs. simulated
`fig_catalog_updated.png`	Updated $\bar{\Lambda}$ ranking with stealthy fixes
`fig_metallic_convergence.png`	$\bar{\Lambda}(\mu_n)$ for $n = 1 \ldots 20$ converging to 1/3
`fig_4x4_alpha_dist.png`	4×4 matrix search: α distribution
`fig_nonunimodular.png`	$\sigma^2(R)$ and running $\bar{\Lambda}$ for $\alpha = 2.071$ chain

Scripts

Script	Purpose
`substitution_tilings.py`	All chains (metallic means, BT, cubic α=2, non-unimodular)
`quasicrystal_variance.py`	$\sigma^2(R)$, $\bar{\Lambda}$, generalized metrics
`projection_method.py`	Cut-and-project Fibonacci chain
`two_phase_media.py`	Spectral density, spreadability
`disordered_patterns.py`	URL, stealthy generation, exact $\bar{\Lambda}$ for URL
`stealthy_analysis.py`	Collaborator stealthy ensemble analysis
`research_catalog.py`	Full pipeline $\rightarrow$ `catalog.json`
`make_figures.py`	Figures from catalog + stealthy results
`analyze_bombieri_taylor.py`	BT-specific analysis
`bt_deep_dive.py`	(Mtg 2–3) BT oscillations, $S(k)$ density, $E(t)$
`finite_size_scaling.py`	(Mtg 2–3) $\bar{\Lambda}(N)$, $C_{II}(N)$, $\hat{\alpha}(N)$
`finite_size_scaling_landscape.py`	(Mtg 2–3) Landscape 1×4 version
`search_cubic_tilings.py`	(Mtg 2–3) Exhaustive 3×3 matrix search
`generalized_metrics.py`	(Mtg 2–3) C_{II}, A_{III}; patches variance module
`finite_size_lambda_bar.py`	(Mtg 2–3) Lattice formula; stealthy two phases
`catalog_cleanup.py`	(Mtg 2–3) Stealthy re-classification, density invariance
`metallic_mean_convergence.py`	(Mtg 2–3) $\bar{\Lambda}(\mu_n)$ for $n = 1 \ldots 20$
`search_4x4_tilings.py`	(Mtg 2–3) 4×4 matrix search (24M matrices)
`fig_nonunimodular_chain.py`	(Mtg 2–3) Figure for $\alpha = 2.071$ non-unimodular chain

Presentations and Written Work

Document	Location	Notes
Week 1 presentation	`results/week1/week1_presentation.tex`	Beamer, core metallic-mean results
Week 2 presentation	`results/week2/week2_presentation.tex`	Beamer, 12 slides
Week 3 presentation	`results/week3/week3_presentation.tex`	Beamer, 16 slides (rewritten 2026-03-24)
Junior Paper	`jp/jp_hyperuniformity.tex`	9 pages, §1–§3 complete
JP results draft	`jp/jp_results_draft.tex`	§4–§8 + appendix, held for later

All documents compile via MiKTeX (`C:/Users/minec/AppData/Local/Programs/MiKTeX/miktex/bin/x64/`).

7 Key Theoretical Results

1. $\alpha = 3$ **universality of metallic means.** For any 2×2 metallic-mean matrix with $|\det M| = 1$: $|\lambda_2| = 1/\lambda_1 \Rightarrow \alpha = 3$ universally.

2. **Bombieri-Taylor fills $1 < \alpha < 2$.** The 3-letter cubic substitution $a \to aac, b \to ac, c \to b$ gives $\alpha = 1.545$, the first known 1D quasicrystal with $\alpha \in (1, 2)$.

3. **First deterministic $\alpha = 2$ substitution tiling.** The chain $c \to abc, a \to bc, b \to c$ gives $\alpha = 2.000$ exactly, $\bar{\Lambda} = 0.275$.

4. **$\bar{\Lambda}$ density invariance.** $\bar{\Lambda}$ is invariant under uniform density rescaling in $d = 1$; direct comparison across patterns with different densities is valid.

5. **Integer lattice has no finite-N correction.** $\bar{\Lambda} = 1/6$ exactly for all integer $R_{\max}$ (piecewise cubic integral identity).

6. **Stealthy two-phase structure.** Both ordered and disordered stealthy phases satisfy $S(k) = 0$ for $|k| < K^*$, but differ drastically in $\bar{\Lambda}$ ($\approx 1/6$ vs. $1/(\pi^2\chi)$). Stealthy α is not a power-law exponent.

7. **$2 < \alpha < 3$ gap is inaccessible to unimodular matrices.** Exhaustive search over 3×3 (39,304) and 4×4 (24M) matrices confirms zero unimodular cases with $\alpha \in (2, 3)$. Rank-3 proof: $|\det M| = 1 \Rightarrow \alpha = 2$ exactly for complex pairs. Rank-≥ 4 proof: additional Pisot factor forces $\alpha < 2$.

8. **General 2×2 formula.** For any 2×2 Pisot matrix: $\alpha = 3 - 2\ln|\det M|/\ln\lambda_1$. Gap $(2, 3)$ is accessible when $|\det M| \in (1, \sqrt{\lambda_1})$; all 28 known examples have $|\det M| = 2$.

9. **First 1D quasicrystal with $\alpha \in (2, 3)$.** $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ ($\det = -2$) gives $\alpha = 2.071$, $\bar{\Lambda} = 0.650 \pm 0.015$. Large log-periodic oscillations in $\sigma^2(R)$, period $\ln\lambda_1 \approx 1.493$.

10. **Silver $\bar{\Lambda} = 1/4$ exact (numerical).** $N = 3.88 \times 10^6$, 3×10^4 windows: $\bar{\Lambda} = 0.2501 \pm 0.0004$ (0.3σ from $1/4$). Analytical proof unknown.

11. **$\bar{\Lambda}(\mu_n) \to 1/3$ confirmed.** Metallic-mean sequence $n = 1 \ldots 20$ converges to $1/3$ at rate $0.22/n^{1.43}$. URL cloaked lattice ($a = 1$, $\bar{\Lambda} = 1/3$) is the exact limit.

8 Open Questions

1. $2 < \alpha < 3$ **gap:** ***RESOLVED*** **(mechanism and first example).** Non-unimodular $M = \left[\begin{smallmatrix} 0 & 1 \\ 2 & 4 \end{smallmatrix}\right]$ gives $\alpha = 2.071$. Sub-question: what is the minimal-complexity chain in $(2, 3)$? (Row sum ≥ 6 is currently the minimum requirement.)

2. **Silver exact value:** ***RESOLVED numerically.*** $\bar{\Lambda} = 1/4$ at 0.3σ. Analytical proof not yet known.

3. $\bar{\Lambda}$ **metallic-mean limit:** ***RESOLVED.*** $\bar{\Lambda}(\mu_n) \to 1/3$; rate $0.22/n^{1.43}$; URL cloaked lattice is the exact limit.

4. **Non-unimodular** $\bar{\Lambda}$**:** Is $\bar{\Lambda} = 0.650$ exactly $13/20$? Analytical proof unknown; large oscillation amplitude makes high-precision measurement difficult.

5. **Bronze closed form:** Is $\bar{\Lambda}(\text{Bronze}) = 0.2822$ a clean expression? Closest simple fraction: $11/39 = 0.28205$ (0.18σ, non-obvious).

6. **Stealthy two-phase literature:** Is the ordered/disordered bifurcation discussed in existing stealthy ground-state literature?

7. **2D generalization:** Does the eigenvalue formula $\alpha = 1 - 2\ln|\lambda_2|/\ln\lambda_1$ extend to 2D substitution tilings (Penrose, Ammann-Beenker)?